

Electric submersible pump vibration analysis under several operational conditions for vibration fault differential diagnosis

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ABSTRACT

An Electric Submersible Pump (ESP) vibration analysis was performed, considering different wear states and operational conditions. The pumps were tested with different fluid viscosities, operating points, and speeds to evaluate their vibration behavior, with the aim of providing characteristics for non-invasive differential vibration diagnosis. A specific frequency spectrum estimation method is described, focusing on the extraction of frequency component vibration amplitudes used in the petroleum industry-standard vibration analysis. A time-interval definition procedure was proposed to reduce signal amplitude losses due to variation of synchronous operating frequency. The results indicate that the frequency component peak amplitudes can be more accurately identified during synchronous frequency variation by the proposed method than by typical estimation methods. In this study, an ESP vibration differential diagnosis was achieved by analyzing relations between orders of the synchronous frequency, and by distinguishing that the synchronous component amplitude rose approximately proportionally to the square of the rotating speed; this differentiates unbalance fault from a bent shaft or a misalignment. A correlation matrix analysis is provided to demonstrate that variation in the fluid-temperature difference between the pump intake and discharge is related to the vibration amplitude variation in a pump with a vibration fault.

1. Introduction

Electrical submersible pumps (ESPs) are the second most common artificial lifting method applied worldwide (Liang et al., 2015), deployed in, it is estimated, 150,000 to 200,000 wells (Flatern, 2015). Electrical submersible pumps account for approximately 10% of the world's crude oil production (Takacs, 2017). These systems are ideally suited to pumping high volumes at high pressure, and the petroleum industry applies the method to pump petroleum at a high flow rate in offshore applications.

An electrical submersible pump is composed of several small-diameter centrifugal stages (typically 4–9 in.) that are mounted serially, with an operational frequency range varying from 30 Hz to 60 Hz. In a submerged ESP installation, a multistage centrifugal pump is coupled to a magnetic induction motor using a protector seal assembly filled with an insulating fluid that is heavier than water. The electric motor is cooled by the oil-well fluids that pass through the motor

housing. For subsea systems, the output motor power may be higher than 1000 hp. The structure can reach more than 40 m in length, and the pump can operate in a high-temperature environment, pumping fluids containing abrasives (Minette et al., 2016).

The installation and intervention costs of ESP-based pumping systems are usually higher than those of other elevation methods, particularly in the case of deep-sea, underwater wells (Ribeiro et al., 2005). In addition to the high costs of installation and intervention, faults in this equipment usually cause significant production losses since they typically operate in high-production petroleum wells. A careful evaluation of ESP systems before installation is critical to prevent premature operational failure.

The risk of vibration problems is increased in equipment such as ESP systems due to the difference between its considerable length and its small axial diameter. The factory acceptance evaluation process of the ESP system for fault diagnosis includes an expert vibration analysis of collected accelerometer signals during a test well operation, following

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the recommendations found in the American Petroleum Institute's recommended practice document 11S8 (API RP 11S8). Despite the rigorous tests performed to detect possible faults, ESP systems still have relatively short run-lives (Borling et al., 2008; Childs et al., 2014), and vibration is the leading cause of failure (Bremmer et al., 2006; Durham et al., 1990; Yao et al., 2011).

The API RP 11S8 does not describe a specific method for estimating frequency component amplitudes in the vibration signals. Therefore, technicians apply different spectrum estimation methods, with different accuracies. Conventional methods consider the vibration signals as stationary, but variation of the synchronous operating frequency has been observed in several ESP facility tests with different configurations. The synchronous operating frequency, or simply synchronous frequency, is the shaft speed, which is approximately the motor's magnetic rotating field, with the difference of a slip frequency that varies with the torque load (Randall, 2010). Gas interference or reservoir pressure variation are some of the causes of ESP shaft-speed variation. A signal component frequency variation over time reduces the amplitude accuracy of the frequency spectra (Brandt and Wiley, 2010).

The API RP 11S8 analysis procedure may suggest up to four different faults with the same probability of producing a symptom, so the correct fault diagnosis is achieved by disassembling the equipment and performing an exhaustive evaluation of the many parts to differentiate the possible failures indicated. Electrical Submersible Pump vibration analysis studies are rare, especially using multiple operating conditions and fluids. In Rauber et al. (2013), for example, a system for the detection and diagnosis of faults ESP systems using a dataset composed of accelerometer sensor data examples from ESP string tests labeled with only three different categories of faults: normal, unbalance or misalignment. The data examples were labeled by visual inspection of the frequency spectra obtained from the Fourier transform of the raw vibration signal. The system used conventional Fourier analysis, which includes frequency spectrum estimation, to extract the root mean square (RMS) amplitude from frequencies from first and second orders (harmonics) of the shaft speed. The results showed an accuracy of around 80%. In Boldt et al. (2014), a performance analysis of automatic classifiers was presented, using a dataset composed of accelerometer sensor data examples from ESP string tests labeled with one of two categories: normal and unbalance. The study used Fourier analysis to extract the amplitude from frequencies of six different orders (harmonics) of the shaft speed. The results showed an accuracy of 98.5%. In Rauber et al. (2017), a comparative study of classifier architectures for automatic fault diagnosis was presented, using a dataset composed of accelerometer sensor data examples from ESP string tests labeled with one of four categories: normal, unbalance, misalignment, and motor rubbing. The study used Fourier analysis to extract the frequency spectra, and statistical features were computed from it. The results showed an accuracy of around 80%. These cited studies do not mention concerns about synchronous frequency variation or strategies to reduce amplitude loss in the frequency spectrum estimation. The results of these studies indicate that the accuracy decreased with more vibration-fault categories, even using different advanced machine-learning methods and architectures. The studies have in common that the labeling process was mainly oriented by the industry standard.

Some of the faults covered by the API RP 11S8 analysis process could be differentiated according to the relationships between synchronous frequency orders' amplitudes. For example, following the API procedure only, high vibrations at the synchronous frequency (1X) indicate unbalance, bent shaft, and misalignment faults. However, usually the bent shaft and misalignment faults produce high vibrations in the 2X order synchronous frequency also (Adams, 2010; Betta et al., 2002; Patel and Darpe, 2009; Scheffer and Girdhar, 2004). Observing the relationship between frequency orders can help the differential diagnosis.

This study aims to investigate ESP vibratory behavior, looking for vibration characteristics beyond the ones observed by the industry's recommend practice, API RP 11S8, that can differentiate the health

condition of pumps and be useful for vibration fault detection and differential vibration diagnosis. Section 2 presents the experimental setup, procedures, and methods of vibration data acquisition following the industry standard's recommendations, with two pumps of the same model but in different unknown health states, seven different operating conditions, five different operating speeds, and two fluids of different viscosities. Fluid flow, pressure, and temperature data were also collected. A frequency spectrum estimation method is proposed in Section 2.3, focusing on extracting frequency component amplitudes observed in the API RP 11S8 procedure analysis and on reducing the amplitude estimation loss due to frequency variation. In Section 3, the ESPs' vibration is analyzed by using the industry-standard analysis on the frequency spectra estimated by the proposed method. The vibration behavior is examined separately against fluid change, rotating speed change, and operating point change. In Section 3.5, a correlation analysis is performed, looking for unanticipated correlations between temperature differences and vibration amplitude.

2. Experimental tests and methods

2.1. Experimental setup

The Artificial Lift Laboratory of the Industrial Automation Technological Training Center at the Federal University of Bahia (UFBA) was used for the ESP tests. The facility was built with industrial equipment to reproduce the petroleum well-field environment in a laboratory. The facility is installed in the midspan of a spiral staircase in the Polytechnic School with 2.5 m of free diameter and 32 m of height, as shown in Fig. 1a. The Artificial Lift Laboratory is supported by Petrobras S/A (Brazilian Petroleum Corporation) and other partners through research and development projects, mainly in the oil-lifting field.

Fig. 1b shows a schematic diagram of the experimental configuration. The ESP system pumps fluid from an artificial well to a tank at the top of the building, and the fluid flows back to the artificial well in a closed circuit. The system was instrumented to control and collect data through a supervisory control and data acquisition (SCADA) system installed in the laboratory control room. Two data acquisition systems receive sensor signal data, one from the accelerometers and another from all other devices.

The ESP system's tubular production string was designed to handle output pressures of up to 1500 psi. Allen-Bradley sensors measure the oil well temperature and pressure at the pump intake and discharge and the fluid flow at the pump discharge. The Allen-Bradley temperature sensor model was 837E-DC1BN1A1D4, the pressure sensor was 836E-DC1ER1D4, and the fluid-flow sensor was 839E-DC1BA1A3D4. Table 1 presents information about these measurement devices.

The ESP vibration acquisition was performed following the recommended practice document 11S8 (API, 2012), using 18 accelerometers orthogonally grouped in pairs, as shown in Fig. 2a. Six accelerometers were installed in each ESP system equipment (ID numbers 1–6 in the motor, 7–12 in the seal, and 13–18 in the pump). All accelerometers were fixed in the ESP sections with a support that was designed and built for the purpose and fastened with a steel belt, as shown in Fig. 2b. The 18 accelerometers were industrial quartz shear ICP® model 624B11 by IMI Sensors with 100 mV/g model sensitivity. Table 2 shows the accelerometers' individual sensitivity from the manufacturer's accredited calibration. During data processing, the individual calibrated sensitivity was used.

The vibration signal acquisition system configuration included a computer, data acquisition hardware, accelerometers, and data acquisition software developed by our research team. The data acquisition hardware connected the accelerometers to the computer. It was composed of a data acquisition chassis and five data acquisition modules installed in the chassis. The chassis model was cDAQ-9172, and the data acquisition modules were NI-9234 models, both from National Instruments (NI). The chassis provided measurements, timing control,

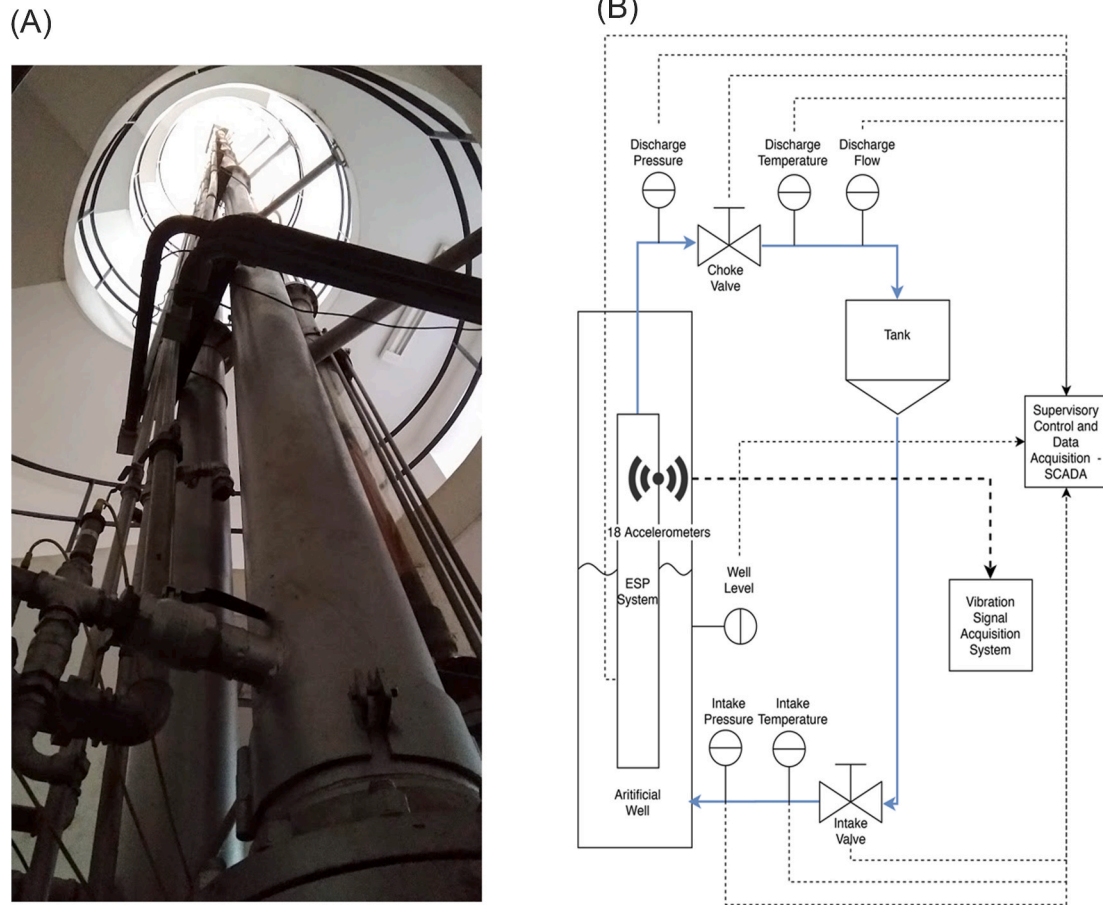


Fig. 1. The experimental facility: a) The artificial oil wells of the Artificial Lift Laboratory installed in the midspan of a 32 m tall spiral staircase in the Polytechnic School at UFBA, b) A schematic of the experimental configuration.

Table 1
Measurement devices.

	Temperature sensor	Pressure sensor	Fluid-flow sensor	Accelerometer
Range	−50 ... +150 °C	0 ... 1500 psi	0.03 ... 3 m/s	±490 m/s ²
Repeatability	<0.1 °C	<0.2%	0.03 m/s	9810 μm/s ² (1–10,000 Hz)
Accuracy	<0.2 °C	<0.5%	±10% on this range	±1%

synchronization, and data transfer between input–output modules and an external computer host. The NI-9234 module was a four-channel dynamic signal acquisition module with 24-bit resolution, up to 51.2 kS/s (where kS is a thousand samples) maximum sampling rate per channel, ±5 V input, 102 dB dynamic range, and antialiasing filters. This project used a 4267 kS/s sampling rate. A customized data-acquisition software was developed in VB.NET language, using the National Instruments Measurement Studio. The software can be used to visualize signals in real-time and to store vibration signals with experiment metadata, such as operating settings (speed, flow/pressure), ESP components used, accelerometer serial numbers, and sensitivities.

Two used Baker Hughes pumps were tested, both 400PMXSSD models, series 400, type P4, with 70 stages. One of the pumps came from an operational oil well and the other from the owner's storage, and they were in different health conditions. The pump model has a recommended flow range from 24 to 95 m³/D. The best efficiency point (BEP) at 3600 RPM operating with water is 72 m³/D with 48% efficiency, with

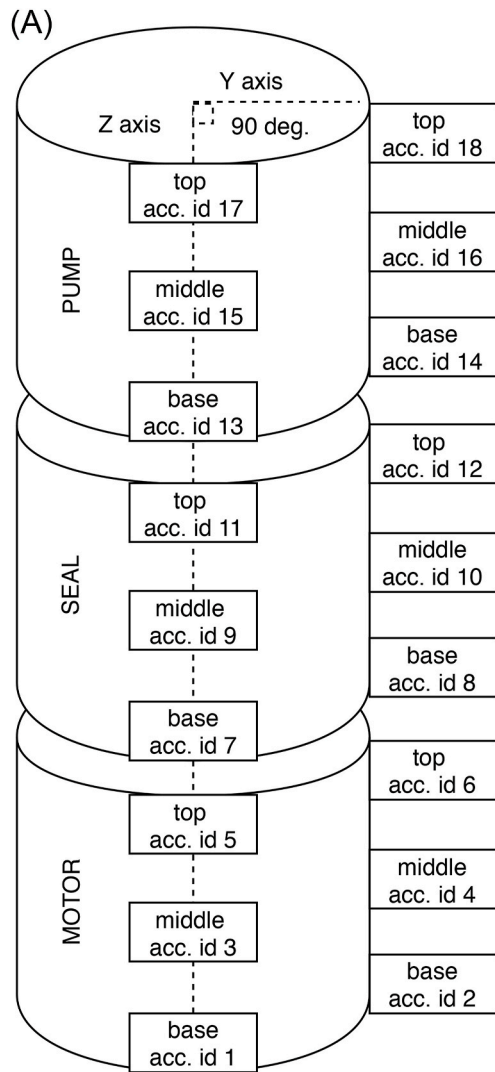
0.22 kW power and 8.8 m head per stage at BEP.

A set of a Baker Hughes components composed of a motor (model FMHX 18HP 465V/25A, Serie 450), a seal chamber (model FSB3XFER SB PFSA HL, Serie 400), and a gas separator (model FRX N AR) were used in all the experiments with both pumps to compare the equipment's vibratory influence. Table 3 presents the geometrical details of the ESP system components.

Each pump was tested with two different viscous fluids. The fluids were produced by Petrobras and formulated from paraffinic petroleum mineral oils. Fluid samples were taken to measure the fluid with a digital viscometer, an SVM 3000 from Anton Paar. To take the fluid samples, the test oil well was fed with each fluid, the recirculating pumping system was operated until it entered a steady state, and then samples were taken. Tables 4 and 5 show the fluids' descriptions: the commercial product name, specific mass, and kinematic viscosity at several tested temperatures. Fluid #2 was a mixture because pure LUBRAX XP 220 is too viscous to be operated by the ESP system used in the experiments.

The operating performance curves of the pump (flow versus pressure at a specific speed) were adjusted to the viscosity of the fluids used in the experiments because the manufacturer's performance curves were based on tests with water at 20 °C and 3560 rpm. As suggested in Borges et al. (2017), Turzo's method (Takacs, 2017) provided the best performance-curve fitting. Thus, this paper uses Turzo's method to obtain the performance-curve corrections. The viscosity at 50 °C was used to correct the manufacturer's performance curves because it was the average pump intake temperature observed during tests under steady-state conditions, as observed by the industry standard and methods for correcting operating performance curves.

To perform the experiments, each pump was installed in the test well



(B)



Fig. 2. The ESP system accelerometer installation schematics: a) Accelerometer distribution between the three main equipment, b) The accelerometer fixing with a support designed and built for the purpose, fastened with a steel belt.

Table 2
Accelerometers used for vibration signal data acquisition.

Acc. ID #	Location	Axis	Calibrated sensitivity (mV/g)
18	Pump Top	Y	99
17	Pump Top	Z	96
16	Pump Middle	Y	99
15	Pump Middle	Z	98
14	Pump Base	Y	98
13	Pump Base	Z	97
12	Seal Top	Y	97
11	Seal Top	Z	98
10	Seal Middle	Y	99
9	Seal Middle	Z	99
8	Seal Base	Y	96
7	Seal Base	Z	98
6	Motor Top	Y	100
5	Motor Top	Z	99
4	Motor Middle	Y	99
3	Motor Middle	Z	96
2	Motor Base	Y	99
1	Motor Base	Z	97

and tested operating with each fluid. Seven different operational points and five different rotational speeds were applied, resulting in 70 different experimental tests per pump with each fluid, a total of 140

experiments. The rotating speeds were 2400, 2700, 3000, 3300, and 3600 rpm (40, 45, 50, 55, and 60 Hz). The operating points were derived from the viscosity-corrected pump-performance curves: the minimum recommended flow (MIN), best efficiency point (BEP), maximum recommended flow (MAX), and four additional intermediate points:

1. MIN
2. 50% of the BEP flow
3. 66% of the BEP flow
4. 83% of the BEP flow
5. BEP
6. 117% of the BEP flow
7. MAX

In each test, the pumping system was started at the preset speed and the flow rate was adjusted by the discharge valve so that the system reached the operating flow point to be tested. After the system entered a steady state, data collection from the measurement sensors was started. Finally, after 160 seconds of steady-state testing, the data collection was terminated, and the test procedures were restarted.

Table 3

Geometrical details of the ESP system components.

	Motor	Seal chamber	Gas separator	Pump
Length	1.37 m	1.71 m	0.81 m	2.14 m
Diameter	11.43 cm	11.43 cm	10.16 cm	10.16 cm
Weight	97 kg	72 kg	34 kg	109 kg

Table 4

Fluid #1: LUBRAX XP 10.

Temperature (°C)	Density (kg/m ³)	Visc. (Cstk)
30	845.9	14.279
35	842.6	12.014
40	839.3	10.212
45	836.0	8.780
50	832.7	7.628
55	829.4	6.682
60	826.0	5.914

2.2. ESP vibration analysis

In the API RP 11S8 document, which includes considerations and definitions of vibration in ESP systems, the potential sources of vibration are listed, for example, as mass unbalance of materials, misalignment of rotating components, flow disturbance from turbulence and cavitation, journal bearing rotation, oil whirl, and mechanical rubbing. This document also presents relations between vibration frequencies to probable causes. These vibration frequencies are orders of the synchronous frequency (X rpm), which is approximately the rotation speed. For example, in the API RP 11S8 document, vibrations at 2X frequency (twice the synchronous frequency), are related to bent shaft fault and misalignment fault as probable causes. In another example, vibrations at 1/2X frequency (half the synchronous frequency), are related to mechanical rub fault and bearing rotating fault as probable causes.

Severe vibration can decrease the ESP system's run-life, as with any rotating machinery. For all ESP components, the API recommends a maximum vibration velocity of 0.156 in/s peak amplitude for each synchronous frequency (1X rpm) and 0.100 in/s peak amplitude for all other frequency components, measured in the chassis of the equipment. This limit must be applied for operation in the manufacturer's recommended ranges of flow, pressure, and speed.

Operation at critical speeds is not recommended, so the vibration limit analysis must account for this possibility. Resonance occurs when the vibration frequency matches a natural frequency of the support system, amplifying the vibration; a critical speed is one that causes this phenomenon. There are some cases of critical speeds found in the range of operation speeds recommended by the manufacturer (Castillo et al., 2019; Minette et al., 2016). Usually, the vibration increases proportionally to the operating speed. A critical speed is a possibility when there is a sudden increase in the vibration amplitude at a specific speed followed by a drastic reduction at a higher speed.

2.3. Frequency spectrum estimation

Typically, the vibration frequency spectrum estimation of rotating

Table 5

Fluid #2: A mix of 67% LUBRAX XP 10 and 33% XP 220.

Temperature (°C)	Density (kg/m ³)	Visc. (Cstk)
30	858.8	29.545
35	855.6	24.089
40	852.3	19.928
45	849.1	16.701
50	845.8	14.150
55	842.5	12.129
60	839.3	10.489

machines operating at a fixed speed considers the vibration signals to be stationary. In this way, the linear spectrum estimator is used with the highest signal length acquired. This ensures the highest possible frequency resolution to guarantee the maximum distinction between vibration phenomena. The linear spectrum estimator is created by applying the discrete Fourier transform (DFT) to the signal multiplied by a window function of the analyst's choice. However, synchronous operating frequency variation was observed in ESP experiments in different experimental configurations and laboratories. The variation may be caused by, for example, gas interference or variations in reservoir level. Whatever the cause, signal frequency variation may reduce the amplitude accuracy of the frequency spectra (Brandt, 2011). The signal frequency variation observed in the experiments is not easy to detect and is small relative to transient signals, but it is enough to reduce the amplitude accuracy, as will be demonstrated in the Results section.

To deal with signal frequency variation, this paper proposes to reduce the measurement time window to the minimum necessary to identify signal components. The reduction of the time window reduces the spectral frequency resolution, resulting in possible difficulties in identifying signal components; in this case, the necessary minimum frequency resolution must be estimated to find the minimum time window.

Spectral frequency resolution is a consequence of the frequency increment and the use of time-window functions. The frequency increment is the distance between two frequency values in the frequency spectrum, which is $\Delta f = 1/T$, so it depends only on the measurement time window, T. The use of time-window functions offers better frequency amplitude representation but causes an energy distribution of the peak amplitude by the size of the window's main lobe; thus, two closed-signal frequency components may superpose each other. The minimum frequency resolution needed to distinguish all API RP 11S8 frequency components was estimated using equation (1).

$$f_{r_{min}} = \frac{\Delta X_{min} f_{o_{min}}}{4L_m} \quad (1)$$

where ΔX_{min} is the minimum difference between API RP 118 synchronous frequency orders (X), $f_{o_{min}}$ is the minimum operating frequency tested, L_m is the width of the main lobe of the window function used, and four is a scalar used to guarantee a minimum distance of two free frequency increments on both sides of each API RP 118 frequency order. The minimum time window needed is

$$T_{min} = \frac{1}{f_{r_{min}}} \quad (2)$$

In this study, the minimum operating frequency was 40 Hz and the Hanning window was used, which has a main lobe width of two. The closest API RP 11S8 frequency orders are 0.48X and 0.5X, and its respective minimum difference is 0.02. From equations (1) and (2) an estimate of a 10-second minimum time interval was obtained.

With the minimum time interval T_{min} determined, the accelerometer signals were processed according to the schematic diagram shown in Fig. 3 to obtain their frequency spectra.

In the data processing, first, the signals are converted from volt units (V) to acceleration units (m/s²) according to the individual calibrated sensitivity of each accelerometer, as shown in Table 2, and then converted from the metric system to the imperial system, which is the system used by the API RP 11S8. The direct current (DC) voltage level was removed by subtracting the mean of each signal.

The trapezoidal integration was used to convert acceleration units to velocity units (in/s) as the industry-recommended practice uses velocity units to evaluate the vibration. A Butterworth IIR high-pass filter of order two with a cut-off of 10 Hz was applied, pre- and post-integration, to eliminate low-frequency anomalies common to the integration process.

The truncation required to convert the signal into a finite length

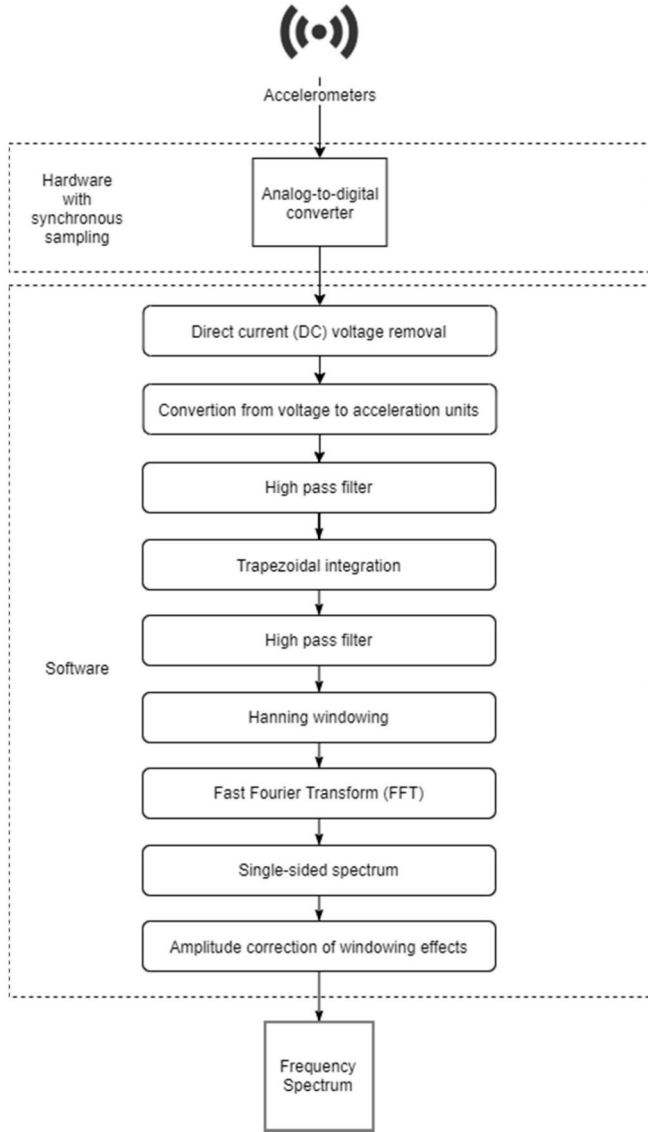


Fig. 3. The procedure used to obtain the accelerometer signal frequency spectra.

causes a leakage effect, which leads to an amplitude error in the frequency spectrum. The signals were processed with the Hanning window to reduce the leakage effect. The Hanning window balances noise injection and reduced amplitude inaccuracy. The Hanning window has a main lobe of two frequency increments, which was used to calculate the minimum time window above. The coefficients of a Hanning window are

$$w(n) = 0.5 \left(1 - \cos \left(2\pi \frac{n}{N} \right) \right), \quad 0 \leq n \leq N-1 \quad (3)$$

where N is the length of the window and n is the index of each coefficient. The Hanning-windowed signal was then the multiplication of the samples $x(n)$ of sampled signal x with the coefficients of the Hanning window

$$x_w(n) = w(n)x(n) \quad (4)$$

For the spectral content of the signals, the DFT was calculated for each signal by

$$X(k) = \frac{1}{N} \sum_{n=0}^{N-1} x_w(n) e^{-j2\pi \frac{kn}{N}}, \quad 0 \leq k \leq N-1 \quad (5)$$

where $x_w(n)$ are samples of a Hanning-windowed signal with N samples, and $X(k)$ are the DFT bins of $x(n)$, where k are indexes of the DFT bins, and i is the imaginary number. The discrete frequencies in the DFT are

$$f(k) = k\Delta f \quad (6)$$

The computation of DFT is efficiently carried out by an Fast Fourier Transform (FFT) algorithm (Oppenheim, 1999).

In computing the frequency spectrum, the single-sided spectrum is adopted, as is shown in the following formula:

$$X_{ss}(k) = 2X(k), \quad 0 \leq k \leq \frac{N}{2} + 1 \quad (7)$$

The windowing process removes energy from the signal, proportional to the sum of the window-function coefficients used, and to compensate for this is necessary to use A_w as a scalar amplitude correction factor (Brandt, 2011)

$$A_w = \frac{N}{\sum_{n=0}^{N-1} w(n)} \quad (8)$$

which, for the Hanning window is precisely two. Then, the final spectrum is

$$X_w(k) = A_w X_{ss}(k) \quad (9)$$

For the peak amplitude extraction of the API RP 11S8 frequency orders, the algorithm first identifies the synchronous frequency by looking for the highest peak amplitude in a 10% range around the equipment's operating speed (with the range accounting for motor frequency slip). With the synchronous frequency, the peaks of the API RP 11S8 orders are identified by searching for the highest peak amplitude within a range determined by twice the frequency increment around the synchronous frequency multiplied by the values of the API RP 11S8 orders (i.e., 1X, 2X, 0.42X to 0.48X).

2.4. Spectrogram

The spectrogram is typically obtained by applying a fixed-length moving window to the time-varying data sequence before computing the spectrum, which is the short-time Fourier transform. Its time-window length is chosen by considering the trade-off between the time and frequency resolutions obtainable (Lee and Han, 1998).

In this study, the typical process to generate spectrograms was applied to the raw accelerometer signal, using five times the samples of the sampling rate (4267×5) as the window size (5 seconds), and an overlap of 200 samples between windows.

2.5. Correlation analysis

Correlation analysis is a method for measuring the covariance of two random variables in a matched dataset. Covariance is typically expressed as the correlation coefficient of two variables, X and Y . A correlation coefficient is a unitless number that varies from -1 to $+1$. The sign of the correlation coefficient shows if the correlation is positive or negative, and the magnitude shows the correlation force between X and Y (De Lima et al., 2009). A negative coefficient indicates that as one variable increases, the other decreases; a positive coefficient indicates that as one variable increases, the other also increases.

Pearson's linear correlation coefficient, one of the most frequently used linear correlation coefficients, was used here.

Pearson's linear correlation coefficient $p(x, y)$ is defined as

$$p(x, y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \cdot \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} \quad (10)$$

where x_i and y_i are samples of x and y indexed with i , $\bar{x} = \sum_{i=1}^n (x_i)/n$, and

$$\bar{y} = \sum_{i=1}^n (y_i)/n.$$

The correlation matrix of two variables is the matrix of correlation coefficients for each pairwise variable combination:

$$R = \begin{pmatrix} p(x,x) & p(x,y) \\ p(y,x) & p(y,y) \end{pmatrix} \quad (11)$$

Since x and y are always directly correlated to themselves, the diagonal entries resolve to one:

$$R = \begin{pmatrix} 1 & p(x,y) \\ p(y,x) & 1 \end{pmatrix} \quad (12)$$

In this study, a correlation matrix was created using the seven sample values of variables from the experiments with seven different operating points, but at the same speed of 3600 rpm. The variables used in the matrix are well-intake fluid temperature, well-discharge fluid temperature, intake–discharge temperature difference, Z-axis accelerometer 1X order peak vibration amplitude at the top of the pump, and well-fluid flow discharge.

3. Results and discussion

In Section 3.1, the amplitude accuracy of the frequency spectrum estimation method is compared to the typical method applied to the quasi-stationary vibration signals acquired in the experiments. In Section 3.2, the vibration of the pumps is analyzed using the industry-standard process, which leads to various possible diagnostics. Sections 3.3, 3.4, and 3.5 present the search for vibration patterns beyond the typical ESP vibration analysis to provide a non-invasive fault differential diagnosis. In Section 3.3, the vibration behavior at different accelerometer positions and in the tests with different fluids is analyzed. In Section 3.4, the vibration behavior in tests at different speeds is analyzed. Finally, in Section 3.5, the vibration behavior in tests of different operating points is analyzed.

3.1. The frequency spectrum of the time-domain vibration signals

Time-domain signal stationarity is essential to the accuracy of frequency spectrum estimation, so spectrograms were used to evaluate the behavior of signal components in time before performing the main

analyses. Fig. 4 shows spectrograms of 160-second signals from the Z-axis accelerometer on top of Pumps #1 and #2, operating with Fluid #1 at 3600 rpm in the BEP. The highest magnitude is in yellow and shows the synchronous frequency vibration of the 160-second accelerometer signal. The synchronous frequency of Pump #1 varied from 56.45 Hz to 56.35 Hz, and that of Pump #2 varied from 56.75 Hz to 56.65 Hz. The slight frequency variation observed, shown in Fig. 4a and b, is probably due to fluid-temperature and viscosity variation during the data collection. This change in frequency may be relatively small, but it reduces the amplitude accuracy of a high-resolution frequency spectrum, which may impair a vibration analysis based on peak limits.

As shown in Section 2.3, equation (1) and equation (2) were used to obtain an estimate of a 10-second minimum time interval to identify API RP 11S8 frequency orders and to minimize frequency variation over time. Fig. 5 shows the frequency spectra obtained from a 160-second time signal and a 10-second time signal from the Z-axis accelerometer fixed on top of Pumps #1 and #2, operating with Fluid #1 at 3600 rpm at the BEP. The figure shows that API RP 11S8 synchronous operating frequency (X) orders were correctly identified by the described method of frequency-spectra estimation and amplitude extraction using both time window intervals. Fig. 5a is the frequency spectrum of the 160-second time signal from Pump #1, the 1X amplitude peak of which was 0.05 in/s; the corresponding peak in the 10-second time signal was 0.08 in/s (Figs. 5c), 60% higher. In the frequency spectra from Pump #2, the 160-second time-signal 1X peak was 0.140 in/s (Fig. 5b), which was below the API RP 11S8 limit, but when using the 10-second time window, the evaluated 1X peak amplitude was 0.313 in/s (Figs. 5d), 120% higher. This reveals that the vibration in this pump was above the API RP 11S8 limit. The other API RP 11S8 frequency orders showed similar differences in amplitude using the shorter time signal.

The results show that the minimum time-signal interval presents a greater amplitude accuracy; thus, this was the time-signal interval chosen for vibration analysis in all experiments.

3.2. API RP 11S8 vibration diagnosis of the pumps

Table 6 shows the vibration peaks of each API RP 11S8 synchronous frequency order during the two pump experiments at 3600 rpm at the BEP with Fluid #1.

Pump #1 did not show amplitudes above the vibration limits in any

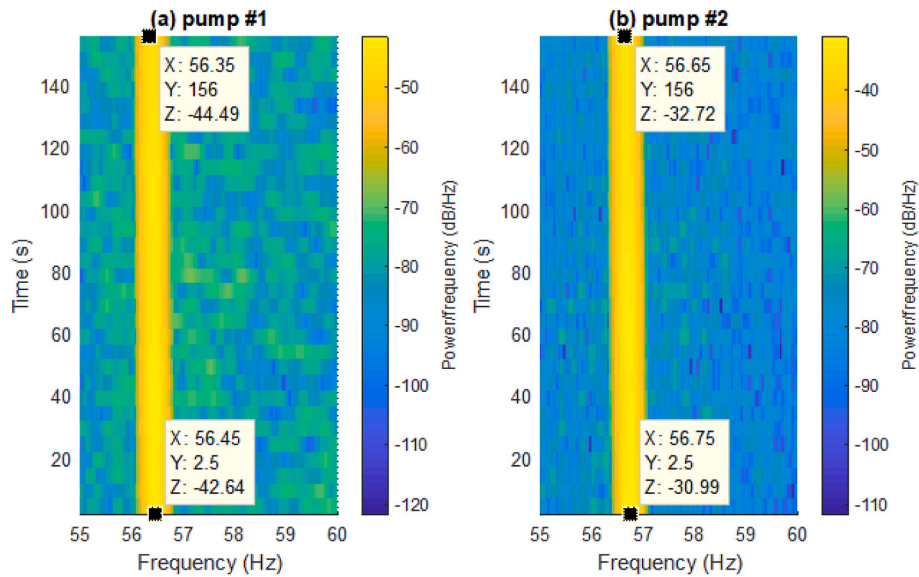


Fig. 4. Spectrograms of 160-second signals from the Z-axis accelerometer on top of Pumps 1# and #2, operating with Fluid #1 at 3600 rpm at the BEP. The highest magnitude is marked in yellow and shows (a) the synchronous frequency in Pump #1 varying from 56.45 Hz to 56.35 Hz and (b) the synchronous frequency in Pump #2 varying from 56.75 Hz to 56.65 Hz. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

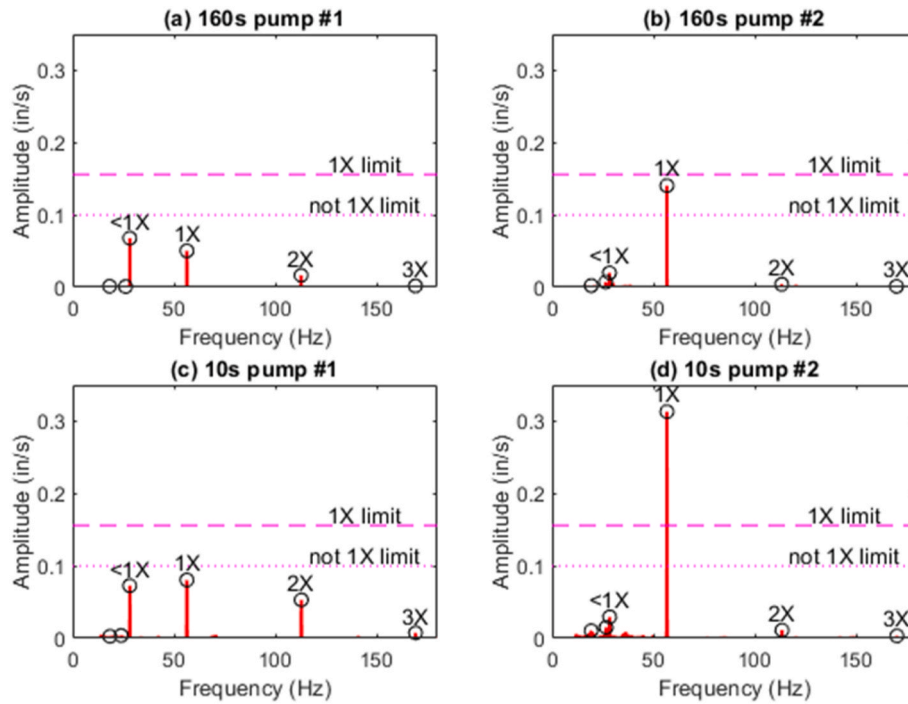


Fig. 5. Frequency spectra from the Z-axis accelerometer on top of the two pumps operating with Fluid #1 at 3600 rpm at the BEP: (a) 160-second signal from Pump #1, (b) 160-second signal from Pump #2, (c) 10-second signal from Pump #1, (d) 10-second signal from Pump #2.

Table 6

Pump accelerometer signal synchronous frequency (X) order peak amplitudes (in/s) at 3600 rpm at the BEP with Fluid #1. Orders were observed according to industry-recommended practice 11S8. Bold case peak amplitudes are above the 11S8 vibration limit.

	Acc. Id.	Location	Axis	X/3	0.42X to 0.48X	X/2	1X	2X	3X	>5X
Pump #1	18	Top	Y	0.003	0.003	0.042	0.053	0.036	0.006	0.003
	17	Top	Z	0.003	0.004	0.073	0.081	0.053	0.008	0.005
	16	Middle	Y	0.002	0.003	0.015	0.060	0.029	0.007	0.006
	15	Middle	Z	0.002	0.004	0.027	0.103	0.041	0.008	0.011
	14	Base	Y	0.004	0.002	0.035	0.078	0.028	0.010	0.003
	13	Base	Z	0.004	0.005	0.066	0.095	0.035	0.012	0.002
Pump #2	18	Top	Y	0.005	0.014	0.028	0.233	0.011	0.002	0.005
	17	Top	Z	0.011	0.015	0.030	0.313	0.011	0.003	0.005
	16	Middle	Y	0.003	0.004	0.017	0.144	0.007	0.003	0.007
	15	Middle	Z	0.004	0.010	0.017	0.165	0.014	0.003	0.004
	14	Base	Y	0.005	0.017	0.036	0.155	0.033	0.004	0.003
	13	Base	Z	0.008	0.018	0.023	0.196	0.043	0.005	0.006

accelerometer. The highest vibration was 0.101 in/s, which was detected by the Z-axis accelerometer at the middle of the pump in the 1X order.

Pump #2 presented several 1X order peaks above the vibration limit, shown in bold in Table 6, especially at the pump top pump, with a maximum of 0.313 in/s. This 1X-order vibration is associated with the following causes according to API RP 11S8:

1. Mass unbalance, hydraulic unbalance, or a decentralized rotor.
2. A bent shaft.
3. A misaligned coupling or shaft bearing or both.

The API RP 11S8 analysis method indicates these three possible diagnostics, which may not necessarily occur simultaneously. Each may involve faults in several internal pump components, which need to be individually inspected for a more precise fault diagnostic.

However, the three diagnostic hypotheses may be further differentiated by analyzing the relationships between the synchronous frequency order amplitudes. Usually, the bent shaft and misalignment diagnostics produce similar high vibrations in the 2X order also, and a

high level of axial vibration (Adams, 2010; Betta et al., 2002; Patel and Darpe, 2009; Scheffer and Girdhar, 2004). In the case of the experiments performed, the axial vibration was not acquired, but as observed in Table 6, the Pump #2 2X amplitude peak was much smaller than the 1X amplitude peak. Although there is a higher possibility of an unbalance fault diagnosis, the 2X amplitude higher peak at the base of Pump #2 and the 1X amplitude higher peak at the top suggest different diagnoses. That is why a better method is needed to differentiate other vibration-behavior characteristics, as described below.

3.3. Vibration at components and fluid change

Fig. 6 and Fig. 7 show the frequency spectra of the accelerometer signals from the ESP system's main equipment (motor, seal, and pump) collected at 3600 rpm at the BEP. Fig. 6a and b are spectra from signals collected during the Pump #1 experiments operating with Fluid #1, and Fig. 6c and d are from experiments with Fluid #2. Fig. 7a and b are spectra from signals collected during the Pump #2 experiments operating with Fluid #1, and Fig. 7c and d are from experiments with Fluid #2. These figures show the vibration peak distribution of all ESP

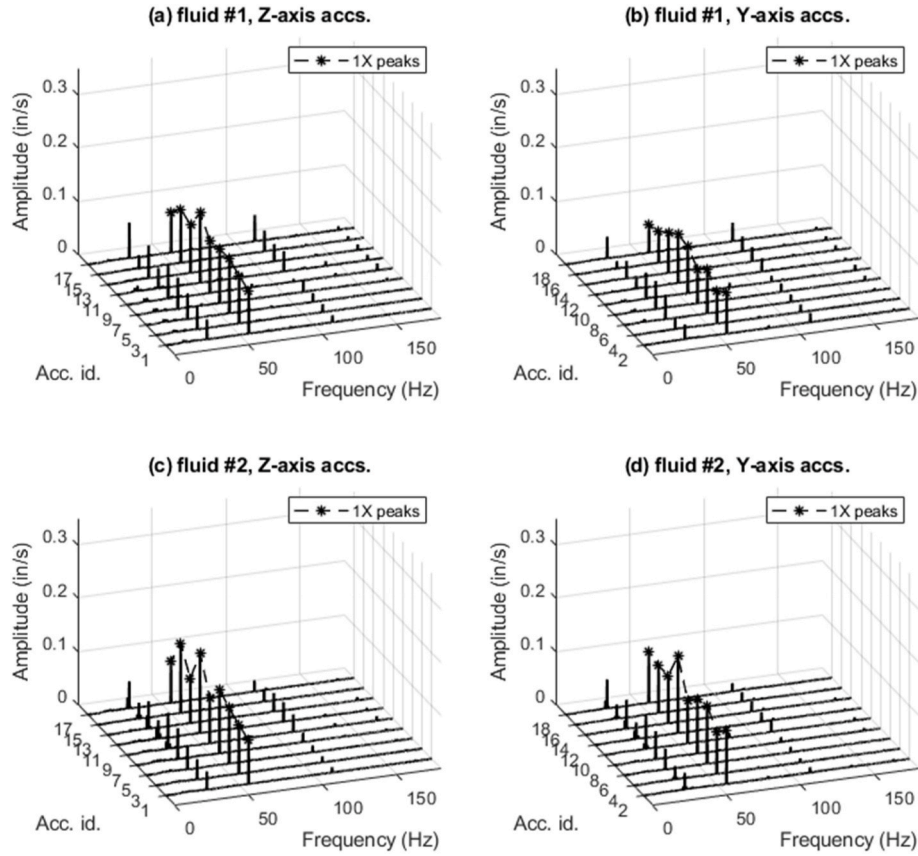


Fig. 6. Frequency spectra of the 18 accelerometers on the Pump #1 ESP system, operating at 3600 rpm at the BEP: a) and b) pumping Fluid #1, c) and d) pumping Fluid #2.

components to allow evaluation of the vibration distribution between components and the influence of the fluid.

In the Pump #1 experiments with Fluid #1, the highest peak was 0.135 in/s from the 1X order of the accelerometer 11 signal (Fig. 6a), which was collected from the seal top at Z-axis. The second highest peak was 0.097 in/s from the 1X order on the Y-axis at the same location (Fig. 6b). These two peaks were below the vibration limit. However, given that the vibration amplitude of neighboring locations exhibited a decreasing behavior, the high peaks at the top of the seal suggest a possible problem in the coupling between the pump and the seal. At the seal top location, the 2X order peak was 0.040 in/s, which is about a third of the 1X order peak at the same location. This proportion suggests a bent shaft or angular misalignment faults in the coupling between the pump and the seal (Adams, 2010; Patel and Darpe, 2009), however the vibration amplitude is below the API RP 11S8 vibration limit.

The highest peak 1X order of ESP for Pump #1 operating with Fluid #2 was 0.155 in/s, registered by accelerometer 11 (Fig. 6c), located on the seal at the top Z-axis, and the second-highest 1X order peak was 0.152 in/s registered on the Y-axis at the same location (Fig. 6d). The overall vibration distribution was similar to the experiment with Fluid #1, except it showed a 14% higher vibration on the seal top Z-axis, which may indicate that the higher viscosity of Fluid #2 increased the vibration.

In the experiment with Pump #2 and Fluid #1, the 1X order highest peak was 0.326 in/s, registered by accelerometer 11 (Fig. 7a), which was located on the seal top Z-axis. This reinforces the indication, observed in the Pump #1 tests using the same seal, that there may be a problem in the coupling between the pump and the seal. The second highest peak of the 1X order was 0.313 in/s, registered by accelerometer 17, which was located on the pump top Z-axis (Fig. 7a); given that this vibration peak was far from the seal, its cause is in the pump itself.

The highest 1X order peak during the experiment using Pump #2 and Fluid #2 was 0.187 in/s, registered by accelerometer 11 (Fig. 7c), which was located on the seal top Z-axis; the second highest 1X order peak was 0.186 in/s, registered by accelerometer 17, which was located on the pump top Z-axis (Fig. 7c). The overall vibration distribution was similar to the experiment with Fluid #1, but with a 41% lower vibration, which may indicate that the higher Fluid #2 viscosity reduced the vibration cause.

The overall vibration amplitudes at the motor and seal components, measured by accelerometers 1–12, were different between tests using Pump #1 and Pump #2, although the components were the same in both assemblies. The motor and seal vibrations were higher when operating with Pump #2 than with Pump #1, which may indicate that these vibration results are primarily a consequence of the pump's operational condition.

Comparing the results using the less viscous fluid (Fluid #1) with the results using the more viscous fluid (Fluid #2), on average there was an increase in the Pump #1 peaks. On the other way, operation with a more viscous fluid caused a reduction in the vibration peaks of Pump #2. The results suggest that, in the good-condition pump (#1), an increase in fluid viscosity amplified the vibration, perhaps causing some unbalance due to greater torsional deformation in the ESP shafts. On the other hand, a more viscous fluid may have attenuated the excessive vibration in the faulty pump, perhaps acting as a damper of its fault.

3.4. Speed variation behavior

Fig. 8a and Fig. 8b show the 1X order vibration peak amplitude and its respective frequency on the pump-top, Z-axis accelerometer in the Pump #1 and Pump #2 experiments, operating at five different rotational speeds at the BEP with Fluids #1 and #2.

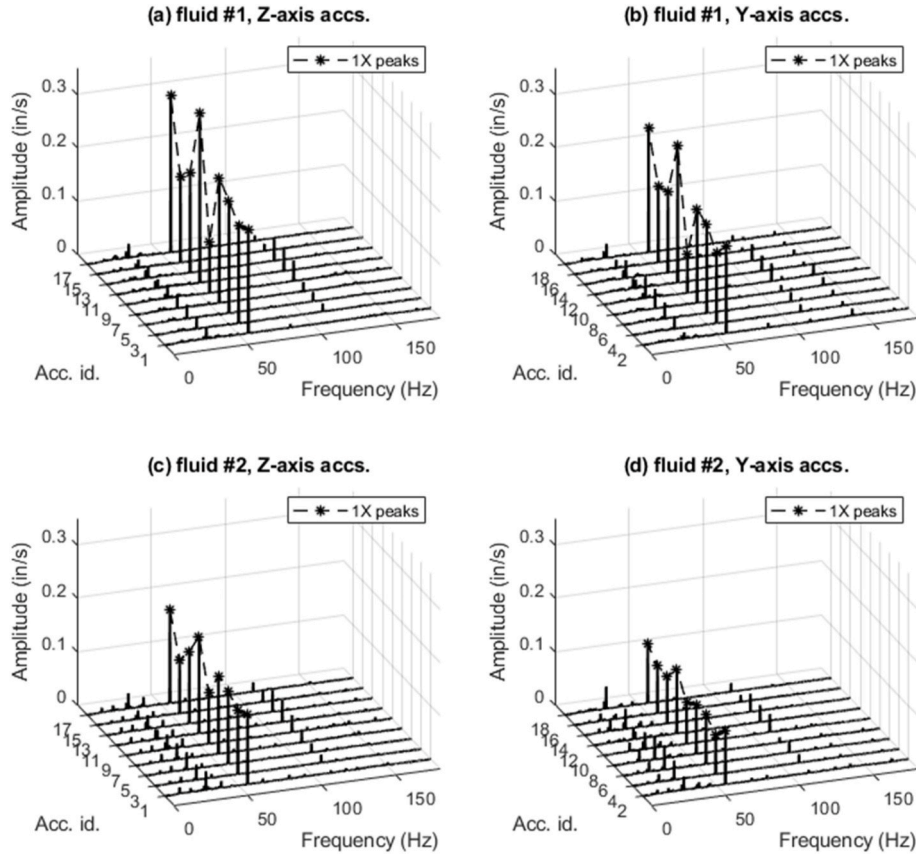


Fig. 7. Frequency spectra of the 18 accelerometers on the Pump #2 ESP system, operating at 3600 rpm at the BEP: a) and b) pumping Fluid #1, c) and d) pumping Fluid #2.

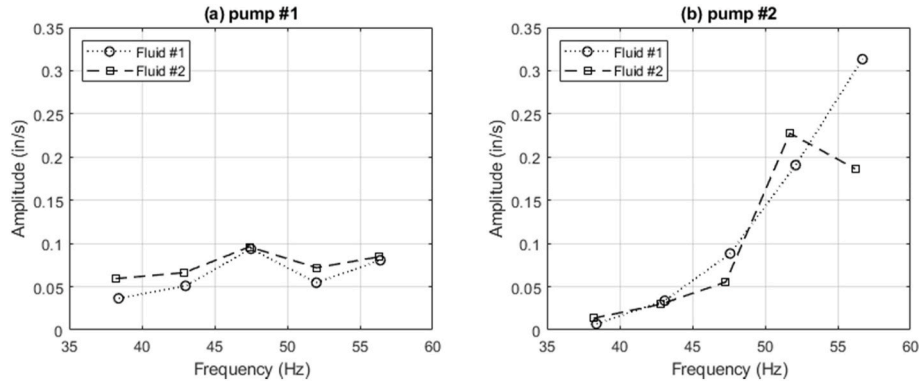


Fig. 8. Z-axis accelerometer 1X order vibration peak amplitudes at the top of the pumps operating at the BEP with Fluids #1 and #2.

An amplified vibration amplitude response was observed in the ESP between 45 Hz and 50 Hz with Pump #1 using both fluids (Fig. 8a), which indicates a resonance condition in Pump #1. Resonance conditions with ESP systems have previously been identified within the manufacturer's operating speed range (Minette et al., 2016). Also, an amplified vibration amplitude response in the ESP was observed between 50 Hz and 55 Hz with Pump #2, but only with Fluid #2 (Fig. 8b). A possible resonance condition in Pump #2, influenced by the properties of Fluid #2, needs to be further examined. Although the two pumps were identical models, they were in different states of wear; the wear state can affect the frequency response of the system, which explains the different resonance-condition possibilities.

The vibration amplitude of Pump #2 at the highest synchronous frequency with both fluids was not associated with a resonance

condition and was above the industry's recommendation of 0.156 in/s (Fig. 8b).

The vibration amplitude of Pump #1 showed a general tendency to increase in proportion to the increase in the synchronous frequency. The vibration amplitude of Pump #2 also showed a general tendency to increase in proportion to the increase in the synchronous frequency, but above the vibration amplitudes of Pump #1.

One of the most common problems that causes vibration is unbalance. As described in Adams (2010), Randall (2016), and Scheffer and Girdhar (2004), the unbalance vibration amplitude varies proportionally to the square of the rotating speed, giving the unbalanced mass centrifugal force, $f = m\omega^2 r$ (where m is the unbalanced mass, ω is the angular velocity, and r is the distance from the unbalanced mass to the center of rotation), is proportional to the square of the rotating speed.

Some degree of unbalance is expected in rotating machines due to material and manufacturing imperfections. However, when the vibration is beyond a limit, the unbalance becomes a problem.

With the Pump #2 vibration peak amplitude above the recommended limit in the 1X order, the industry-recommended practice indicates unbalance, bent shaft, or misalignment as probable causes, either simultaneous or not. However, given the increase in vibration amplitude associated with the increase in synchronous operating frequency, the unbalance vibration diagnosis is more likely.

3.5. Flow and pressure variation behavior

Fig. 9a and Fig. 9b show the vibration peak amplitudes of 1X order at the pump-top Z-axis accelerometer on Pumps #1 and #2 during seven operating points tests at the 3600 rpm speed (60 Hz), pumping Fluids #1 and #2. Fig. 9a shows that varying the operating point had little influence on the vibration amplitude at Pump #1, regardless of the fluid used. In contrast, Fig. 9b shows considerable variation in the vibration amplitude with Pump #2, and the variation curve seems to be without tendency. As the shapes of the diffusers and impellers in Pump #2 may have been modified due to wear, with fluid dynamic consequences, it is not possible to associate a specific physical phenomenon to the great vibration peak variation shown in Fig. 9b. The variation does not seem related only to the changes in the operating point, and the curve is similar using Fluids #1 and #2.

Correlation matrices were generated, as described in Section 2.5, to investigate possible additional effects influencing the Pump #2 amplitudes at several operating points. Fig. 10 shows correlations between pairs of variables measured during Pump #2 experiments operating at 3600 rpm: the well-intake fluid temperature (T_{in}), the well-discharge fluid temperature (T_{dis}), the intake–discharge temperature difference (T_{dif}), the Z-axis accelerometer 1X order peak vibration amplitude at the pump-top (1X peak), and the well-fluid flow discharge (Q_{dis}). Fig. 10 also shows the histograms of the variables, the matrix diagonal, and scatter plots of the variable pairs. The slopes of the least-squares reference lines in the scatter plots are equal to the correlation coefficients.

In the Fig. 10, the -0.17 correlation value between the well-intake temperature (T_{in}) and the 1X order peak amplitude indicates that there was no significant influence between the intake temperature variation and the pump vibration's peak amplitude. The 0.85 correlation value between the well-intake temperature and its discharge temperature (T_{dis}) is high and was expected because well-intake temperatures influence the discharge temperatures. The 0.36 correlation value between the well-discharge temperature and the 1X order peak amplitude indicates that the well-discharge temperature had a more significant influence on the 1X order peak amplitude than the intake temperature. The 0.95 correlation value between the intake–discharge temperature difference (T_{dif}) and the 1X order vibration peak amplitude is higher than any other correlation. This may indicate that the vibration amplitude variation when the operating point was varied is more related to

the fluid temperature difference between the pump intake and discharge than other variables. Neither the intake temperature nor the discharge temperature alone showed such a high correlation with vibration.

Fig. 11a and Fig. 11b show the Z-axis accelerometer 1X order vibration peak amplitudes at the top of Pumps #1 and #2, together with the oil well's intake and discharge fluid temperature difference, by operating point at 3600 rpm. Fig. 11a shows that the 1X order vibration amplitude of Pump #1 did not vary while there was substantial fluid-temperature-difference variation. Fig. 11b shows that the 1X order vibration amplitude of Pump #2 was significantly correlated to the fluid-temperature difference. Fig. 12a and Fig. 12b show the same relationships in the Fluid #2 experiments.

The increased intake–discharge temperature difference in the faulty pump tests was expected because an additional part of the energy given to the fluid is transformed into heat due to the reduced efficiency of used pumps. It is well known that fluid temperature may significantly influence viscosity and may also affect the stiffness of pump components. However, both may not affect vibration in the same way. Since Pump #1 does not present vibration above the industry-standard limit and does not present much amplitude variation under fluid-temperature-difference variation, while Pump #2 shows, this characteristic may be seen as another fault symptom in Pump #2, and that fault appears to be strongly influenced by the temperature difference between the intake and discharge.

The temperature difference in the pump extremes can cause a difference in fluid density sufficient to create an axial unbalance, but, to this axial unbalance affect Pump #2 only, his shaft could be with reduced stiffness. This characteristic may reinforce the diagnostic hypothesis that this pump has unbalance, and the hypothesis that it has reduced stiffness in the shaft. This high correlation between temperature difference and 1X order vibration peak amplitude may also be used as complementary information to help in the differential diagnosis of vibration faults.

4. Conclusions

In this paper, an experimental investigation of the vibration behavior of two ESPs in different wear states under different operating conditions was presented. The experiments followed industry recommendations, using an artificial lift test oil well and viscous fluids to reflect field operational conditions. The pumps were tested with two different viscous fluids, seven different operating points (flow versus pressure), and five different speeds to evaluate their vibration behavior.

A frequency spectrum estimation method was described, focusing on extracting the vibration amplitude from frequency components used in API RP 11S8 analysis and minimizing the amplitude loss in signals with frequency variation. The experimental results indicated that the synchronous frequency varied during the experiments, and the API RP 11S8 frequency components were effectively identified by the frequency estimation method, with more accurate peak amplitudes than those

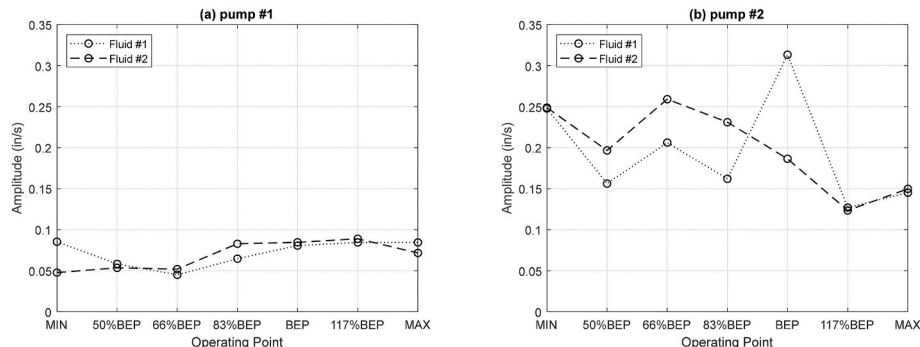


Fig. 9. Z-axis accelerometer 1X order vibration peak amplitudes at the top of the pump by operating point, at the BEP with Fluids #1 and #2.

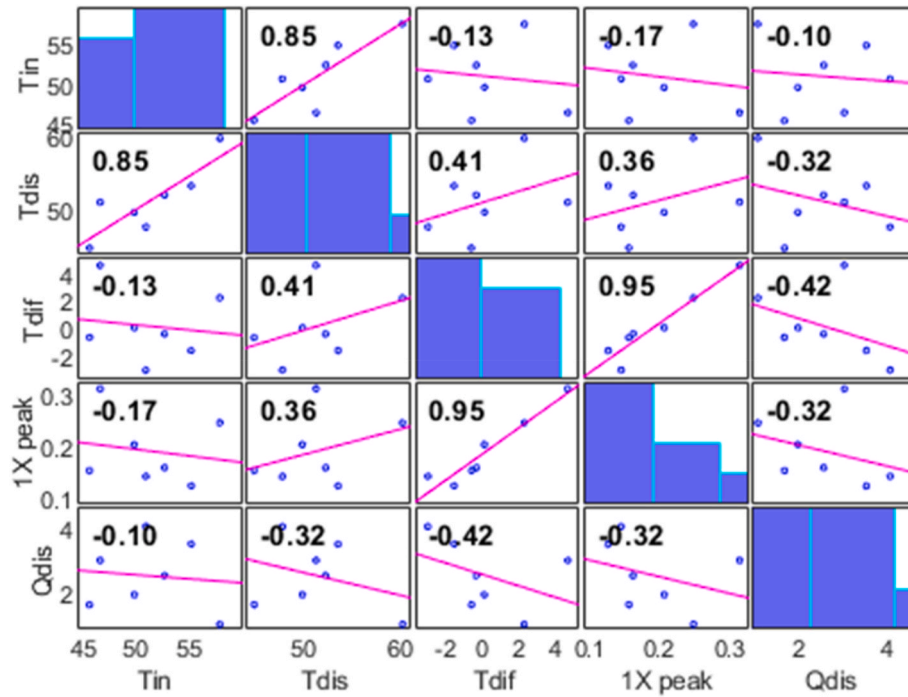


Fig. 10. Experimental variable correlations from Pump #2 operating at 3600 rpm at several operating points: well-intake fluid temperature (T_{in}), well-discharge fluid temperature (T_{dis}), intake–discharge temperature difference (T_{dif}), Z-axis accelerometer 1X order vibration peak amplitude at the top of the pump (1X peak), and well-discharge fluid flow (Q_{dis}).

obtained using a typical method.

The results of the industry-recommended vibration analysis indicated that a faulty pump exceeded the vibration limit by more than double at the synchronous frequency (1X). According to the recommended practice, three possible faults (not necessarily co-occurring) may be suggested: mass unbalance, bent shaft, and misalignment. The API RP 11S8 vibration analysis does not indicate a non-invasive method for diagnostic differentiation. Regardless of the standard industrial analysis, the relationship of peak vibrations between orders of the synchronous frequency indicated a greater likelihood of unbalance.

The vibration behavior analysis with fluid change indicated that the fluid's viscosity had a more significant influence in the pump that showed higher vibration, decreasing the amplitude. The pump that was in good condition, with low vibration, showed slightly vibration amplitude increase with the more viscous fluid but remained far from

the vibration limit. The high-vibration pump showed less vibration amplitude when operating with the more viscous fluid but remained above the vibration limit, which may indicate that the viscosity acted as a damper to the vibration cause.

The vibration behavior analysis at different operating points showed no significant amplitude variation in the pump that was in good condition and high amplitude variation in the faulty pump. Exploratory analysis using a correlation matrix indicated that the vibration amplitude in the faulty pump was highly correlated to the fluid-temperature difference between the well's intake and discharge, while the discharge temperature and intake temperature alone showed lower correlations with the vibration amplitude.

Although the increased intake–discharge temperature difference in the pump tests was expected because an additional part of the energy given to the fluid is transformed into heat due to the reduced efficiency

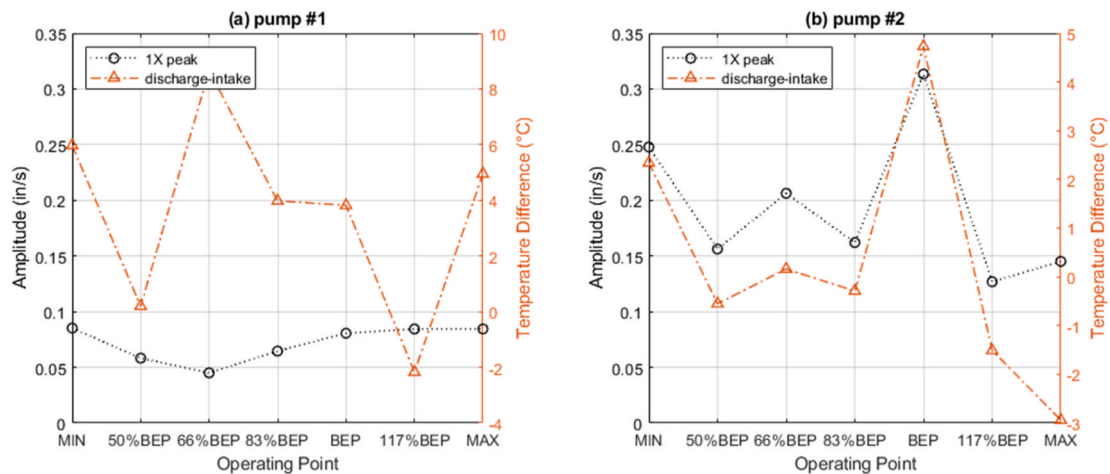


Fig. 11. Z-axis accelerometer 1X order vibration peak amplitudes at the top of the pump and the intake–discharge fluid temperature difference by operating point at 3600 rpm with Fluid #1.

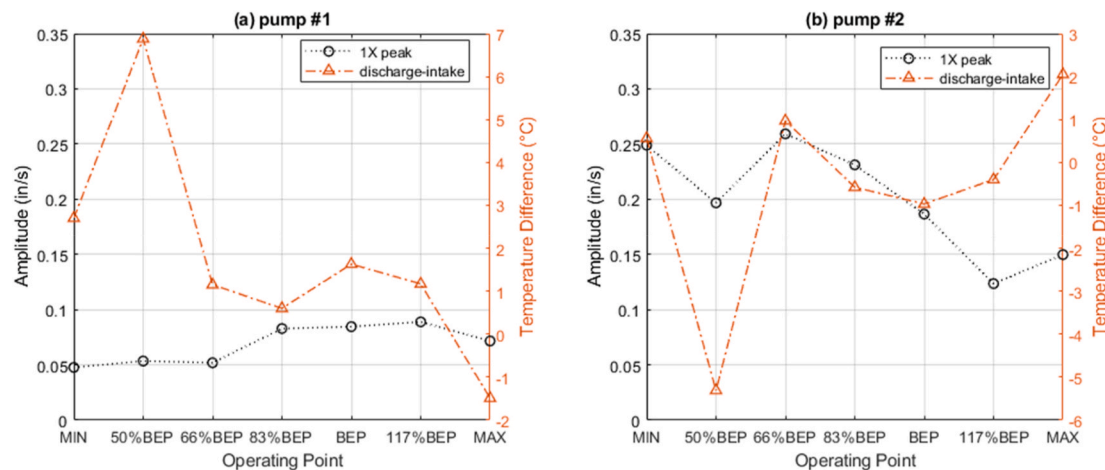


Fig. 12. Z-axis accelerometer 1X order vibration peak amplitudes at the top of the pump and the intake–discharge fluid temperature difference by operating point at 3600 rpm with Fluid #2.

of used pumps, the high correlation between temperature difference and 1X order vibration peak amplitude can be applied as complementary information during vibration diagnosis.

The vibration behavior analysis of different speeds demonstrated that the faulty pump presented a vibration amplitude increase approximately proportional to the square of the speed, which is correlated to an unbalance fault. This characteristic may also be used for ESP differential diagnosis from the other API RP 11S8 fault options involving similar spectra, such as misalignment or a bent shaft.

In future research, it is planned to use more accelerometers, including axial accelerometers, to manipulate fluid temperature, and to cause controlled faults in the pumps.

CRediT authorship contribution statement

Galdir Reges: Conceptualization, Methodology, Software, Investigation, Formal analysis, Visualization, Writing - original draft, Writing - review & editing. **Marcio Fontana:** Conceptualization, Supervision, Methodology, Writing - original draft, Writing - review & editing. **Marcos Ribeiro:** Project administration, Resources, Validation, Writing - review & editing. **Tiago Silva:** Supervision, Data curation. **Odilon Abreu:** Investigation, Data curation. **Ricardo Reis:** Software, Data curation. **Leizer Schnitman:** Project administration, Resources, Validation, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.oceaneng.2020.108249>.

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